

Troubleshooting Guide: Seawater Cooler — Fouling and Runtime Prediction

Category: Heat Transfer Equipment / Predictive Maintenance **Unit:** Seawater-Cooled Shell-and-Tube Heat Exchanger (process gas/liquid cooling service) **Tools:** HTRI Xchanger Suite (thermal-hydraulic rating with fouling-resistance trending) **Fluid Package:** Not applicable in the traditional VLE sense — see Section 4, Step 1 **Symptom:** Gradual multi-month performance decline — not a single failure event

Note on case type: Unlike Case Studies 1–5, this is primarily a **predictive/quantitative trending exercise** rather than a single-root-cause diagnosis. The underlying cause (biological fouling) was reasonably suspected from the start given seawater service; the real engineering task was **quantifying** the degradation and **predicting remaining runtime**, not distinguishing between competing hypotheses. It’s included here because the same “model vs. actual” discipline used elsewhere in this series applies directly to predictive maintenance, not just fault diagnosis.

1. Quick Reference

Item	Detail
Reported symptom	Gradually worsening performance over months: rising process outlet temperature, higher compressor discharge temperature, increasing seawater-side pressure drop, climbing energy use
Task	Quantify the degradation and predict remaining runtime before the exchanger could no longer meet cooling duty
Actual root cause	Biological fouling (algae and marine growth) on the seawater side, confirmed by inspection
Fix	Mechanical cleaning; optimized chlorination/biocide dosing
Diagnostic tool	HTRI Xchanger Suite model; overall U (measured) vs. clean design U; historical fouling-rate trend extrapolated forward
Prevention	Established an ongoing HTRI-based fouling monitoring program — cleaning now scheduled by predicted remaining runtime , not a fixed calendar interval

2. Symptom

A **gradual, multi-month decline** across several related indicators — the pattern itself (slow, compounding, multi-parameter) is the first diagnostic clue:

- **Rising outlet process temperature** (cooling duty not being fully met)

- **Higher compressor discharge temperature** (downstream consequence of reduced cooling)
- **Increasing seawater-side pressure drop** (physical restriction developing)
- **Climbing energy use** (system working harder to compensate)

3. Why This Called for Quantification, Not Just a Cleaning Call

With seawater cooling service, **biological fouling** (algae, marine growth, scaling) is a well-known and expected long-term degradation mechanism — this isn't really a "which of several unrelated causes is it" situation like Cases 1–5. The actual engineering question was different:

1. **How much has performance actually degraded**, in quantitative terms (not just "temperatures are creeping up")?
2. **How much runtime is left** before the exchanger can no longer meet cooling duty?
3. Should cleaning happen **now**, or can it be scheduled to align with a planned outage?

Reacting to each symptom individually (temperature creeping up → schedule a clean) doesn't answer these questions — it requires a **model-based quantitative trend**, not just a qualitative "it's fouling" observation.

4. Diagnostic Approach

Step 1 — Model the exchanger using design geometry and operating data

The exchanger was modeled in **HTRI Xchanger Suite**, using: - Design tube/shell geometry
- Actual operating data (flows, temperatures)

Why there's no "fluid package" here in the usual sense: Unlike Cases 1–5, which all involve phase-equilibrium (VLE/electrolyte/hydrate) modeling requiring a specific thermodynamic package, this is a **thermal-hydraulic rating exercise**. HTRI uses its own built-in physical property correlations and databanks (seawater properties, process-side fluid properties as input) rather than a cubic-EOS flash calculation — there's no vapor-liquid equilibrium being solved. This is a useful distinction to keep in mind: not every simulation-based diagnosis is a VLE problem, and picking the right *type* of tool (rating software vs. flash-based process simulator) matters as much as picking the right fluid package within a simulator.

Step 2 — Calculate actual overall heat transfer coefficient (U) and compare to clean design value

U_{actual} (from measured temperatures and flows) vs. U_{clean} (design/clean condition)

The gap between these two values is a **direct, quantitative measure of fouling resistance** — not just an inference from temperature trends, but a calculated number tied to the underlying heat transfer physics.

Step 3 — Trend the fouling rate historically and extrapolate forward

Using **historical trend data** (not just the current snapshot), the **fouling rate** was estimated — i.e., how quickly U is degrading over time. This rate was then **extrapolated forward** to predict when U would fall below the value required to meet cooling duty.

Interpretation: This converts a reactive question ("performance is bad, should we clean now?") into a proactive one ("at the current fouling rate, we have approximately N weeks/months of acceptable performance left — plan the clean accordingly").

Step 4 — Validate against inspection

The HTRI-based prediction was checked against physical inspection findings to confirm the model's fouling-rate estimate and mechanism assumption were correct, not just numerically self-consistent.

Quantitative Basis

- Clean design U: 210 Btu/hr·ft²·°F. Trended actual U fell from 205 (month 1) to **118 Btu/hr·ft²·°F by month 7 — a 44% loss.**
- Average fouling rate over the trend: approximately 0.00031 hr·ft²·°F/Btu per week.
- Extrapolating the trend forward predicted U would fall below the 140 Btu/hr·ft²·°F minimum-duty threshold within **approximately 3 weeks** of the assessment — the basis for the runtime prediction communicated to operations.

5. Root Cause

Biological fouling — algae and marine growth — on the seawater side, confirmed by inspection and consistent with the fouling-resistance trend calculated from the HTRI model.

6. Corrective Action

1. **Mechanical cleaning** of the exchanger to remove accumulated biological fouling.
2. **Optimized chlorination/biocide dosing** on the seawater side to slow future biological growth.

7. Verification

- Post-cleaning U measured at **208 Btu/hr·ft²·°F — 99% of the 210 clean design value**, confirming fouling (not tube degradation) was the sole driver of the decline.
- Chlorination free residual dosing increased from 0.3 to **0.6 ppm.**
- Re-trended fouling rate over the following 3 months showed a much slower decline; extrapolated runtime at the new dosing rate extended from **3 weeks to over 6 months.**

8. Prevention / Long-Term Fix

- **Established an ongoing HTRI-based fouling monitoring program.**
- Cleaning is now **scheduled based on predicted remaining runtime** (from the U-trend extrapolation), rather than a **fixed calendar interval.**
- This shift — from calendar-based to condition/prediction-based maintenance — improves reliability (catching degradation before it becomes duty-limiting) while also **cutting unnecessary maintenance cost** (not cleaning exchangers that don't yet need it).

9. General Troubleshooting Checklist (Reusable)

Use this checklist for seawater-cooled (or similarly fouling-prone) heat exchanger gradual performance decline:

- ☐ Confirm the decline is gradual/multi-parameter (temperature, dP, energy use) rather than a sudden step change (which would point to a different cause, e.g., tube leak, blockage, instrument fault)

- ☐ Gather design geometry and current operating data (flows, inlet/outlet temperatures both sides)
- ☐ Build/update a thermal-hydraulic rating model (e.g., HTRI Xchanger Suite) for the exchanger
- ☐ Calculate **actual overall U** from measured data and compare to **clean/design U**
- ☐ Pull **historical trend data**, not just a current snapshot, to estimate the fouling rate
- ☐ **Extrapolate forward** to predict remaining runtime before U drops below the value needed for required duty
- ☐ Cross-check the predicted fouling mechanism against physical inspection when the unit is next opened
- ☐ Use the runtime prediction to **schedule** cleaning (aligned with planned outages where possible) rather than reacting to symptoms or defaulting to a fixed calendar interval
- ☐ Review chemical treatment (chlorination/biocide dosing) as a complement to mechanical cleaning, not a replacement for it
- ☐ Set up an **ongoing** monitoring cadence (not a one-time study) so future runtime predictions stay current as conditions change

10. Key Takeaway

When degradation is gradual and expected (like biofouling in seawater service), the value of simulation isn't in *discovering* the cause — it's in **quantifying** it and **predicting** how much time you have left. Calculating actual U against clean design U, then trending and extrapolating that gap, turns “temperatures are creeping up, better clean it soon” into a defensible runtime number you can plan an outage around. That shift — from calendar-based to prediction-based maintenance — is often where the real cost savings and reliability gains come from, not just the cleaning itself.

Related Concepts / Tags

seawater-cooler shell-and-tube fouling biofouling HTRI Xchanger-Suite overall-heat-transfer-coefficient U-value-trending runtime-prediction predictive-maintenance chlorination biocide-dosing thermal-hydraulic-rating

This guide is derived from a real field troubleshooting case. Values and thresholds shown are specific to the reported case and should be validated against your own unit's design basis before applying.